Contents

Getting started

Everything you need for this course runs inside a Docker container: Python, JupyterLab, NumPy, pandas, scikit-learn and matplotlib are already installed. You do not install Python on your machine.

You will never need an API key or a paid account for this course.

The image comes in two sizes. You start with `core`, which is everything lessons 1 to 8 need. Before Lesson 9 you switch one line to `full`, which adds TensorFlow for the two neural-network lessons — and because `full` is built on top of `core`, that switch downloads only the part you do not already have.

---

1. Install Docker Desktop

Download it from docker.com and start it. Windows, macOS and Linux are all fine.

You need **8 GB of RAM**, and about **10 GB of free disk space** — 4 GB for the `core` image and room for the `full` one in November.

---

2. Clone the course repository

```
git clone https://github.com/fabioantonini/technologies-for-artificial-intelligence.git
cd technologies-for-artificial-intelligence
```

This is how you will receive new lessons throughout the course, so clone it somewhere you will find again — not your Downloads folder.

---

3. Start the environment

From inside the repository folder:

```
docker compose pull
```

Do this at home, before the first lecture. It is about **750 MB** over the network, which unpacks to roughly 3 GB on disk. That is a few minutes on a home connection and a bad afternoon if thirty of us start it at once on the lecture-theatre wifi. Then start it:

```
docker compose up
```

Later runs start in seconds and download nothing.

Then open:

```
http://127.0.0.1:8888/lab?token=aicourse
```

JupyterLab opens with the course material already there. Anything you save is written to your own machine, inside the folder you cloned.

To stop it, press `Ctrl+C` in the terminal, or run `docker compose down`.

---

4. Getting each new lesson

New material is published on the day of each lesson. To fetch it:

```
git pull
```

That is all. A few megabytes, a couple of seconds — **you do not re-download the Docker image.** The image only changes when the software environment does, which is rare, and you will be told when it happens.

> **If you have edited a course notebook** and `git pull` complains about a conflict, the simplest fix is to copy your version to a new name (`my_lesson3.ipynb`), then run `git checkout -- .` and pull again. Better still: work on copies from the start, and keep your own work in a `my_work/` folder.

Troubleshooting

Port 8888 is already in use. Something else is using it. Either stop that, or edit `docker-compose.yml` and change `"8888:8888"` to `"8889:8888"`, then open `http://127.0.0.1:8889/lab?token=a`

"Permission denied" when saving a notebook. This should not happen — the compose file is configured to prevent it. If it does, stop the container, run `docker compose down`, then `docker compose up` again. Report it if it persists.

The container is slow to start the first time. Expected: the image is initialising and the environment is large. Subsequent starts are much faster.

I deleted my container, is my work gone? No. Your files live in the folder you cloned, on your own machine. Containers are disposable; your work is not.

Do I need a GPU? No. Every lab in this course runs on CPU by design.

Before Lesson 9: switching to the full image

Lessons 9 and 10 use TensorFlow, which is large and which nothing before them needs. It is therefore not in the image you have been using. To add it, open the `.env` file in the repository folder and change one line:

```
TAI_TAG=full
```

Then, at home and before the lecture:

```
docker compose pull
docker compose up
```

You are not downloading the environment again. Everything except TensorFlow is already on your machine, so only that layer is fetched.

Running without cloning (not recommended)

The published image contains a snapshot of the course, so this works:

```
docker run --rm -p 8888:8888 -e JUPYTER_TOKEN=aicourse \
  fabioantonini/technologies-for-artificial-intelligence:core
```

But your work is stored **inside the container and lost when it stops**, and the snapshot inside the image is frozen at the moment it was built, so you would not receive new lessons at all. Use the `git clone` route above.